

iClusterBayes Summary Report

Automated Report

Hyperparameters

```
cat('sdev = 0.02, beta.var.scale = 0.4\n')
```

```
## sdev = 0.02, beta.var.scale = 0.4
```

Key Summary Statistics

```
cat(paste0('```\n', '

--- Summary for K = 1 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.4678  0.5061  0.5139  0.5139  0.5217  0.5494

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.0000  0.0950  0.1394  0.1359  0.1711  0.8944
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.00000 0.09778 0.14222 0.13881 0.17444 0.72889
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.000000 0.005000 0.008333 0.102034 0.031111 1.000000
```

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Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0005556 0.0050000 0.0083333 0.0780456 0.0238889 1.0000000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0061111 0.0150000 0.162453 0.117778 1.000000

--- Summary for K = 2 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
 0.2128  0.2356  0.2428  0.2427  0.2506  0.2733

beta.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.05278 0.10278 0.10558 0.14222 0.80556
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1

gamma.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.05556 0.10667 0.10858 0.14667 0.66611
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.001667 0.060556 0.416370 1.000000 1.000000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.001111 0.005000 0.250076 0.386250 1.000000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
 0.0000  0.1328  1.0000  0.6989  1.0000  1.0000

--- Summary for K = 3 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
 0.1217  0.1522  0.1606  0.1621  0.1706  0.2417

```

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beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01944 0.05833 0.06923 0.10222 0.89278
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.02000 0.06111 0.07181 0.10667 0.62556
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.7661  1.0000  0.7733  1.0000  1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0000000 0.0005556 0.2259396 0.2123611 1.0000000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.1172  1.0000  0.6837  1.0000  1.0000

--- Summary for K = 4 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.03833 0.06111 0.06833 0.06936 0.07556 0.14000

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01944 0.06000 0.06899 0.10000 0.84833
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :

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Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02111	0.06444	0.07269	0.10667	0.65500

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.5683	0.8311	0.7057	1.0000	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.0000	0.0000	0.2704	0.6675	1.0000

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.9950	1.0000	0.8584	1.0000	1.0000

--- Summary for K = 5 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.03111	0.04944	0.05611	0.05862	0.06556	0.14222

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01556	0.04944	0.06017	0.08778	0.83556

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01667	0.05333	0.06408	0.09444	0.59889

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.2744	0.6322	0.5835	0.9906	1.0000

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.8622	1.0000	1.0000	1.0000	1.0000	1.0000

Modality 4 :					
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    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000  0.0000  0.2524  0.5406  1.0000
Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.8672  1.0000  0.8440  1.0000  1.0000

--- Summary for K = 6 ---
Z.ar:
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.03667 0.05944 0.06889 0.07106 0.07944 0.15111

beta.ar:
Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.005556 0.028889 0.045548 0.067222 0.799444
Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1
Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1
Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1
Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1         1         1         1         1         1

gamma.ar:
Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000000 0.005556 0.031111 0.048393 0.073333 0.595556
Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.08778 0.36773 0.81778 1.00000
Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.6656  1.0000  1.0000  0.9996  1.0000  1.0000
Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.0000  0.0000  0.1860  0.2053  1.0000
Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000  0.4906  0.8444  0.6852  0.9828  1.0000

--- Summary for K = 7 ---
Z.ar:
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.2239  0.2517  0.2622  0.2641  0.2744  0.3517

beta.ar:
Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

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0.00000 0.00500 0.01778 0.03216 0.04222 0.74000
Modality 2 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
Modality 3 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1

gamma.ar:
Modality 1 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.005556 0.020000 0.033246 0.045556 0.497778
Modality 2 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.000000 0.000000 0.13399 0.08167 1.00000
Modality 3 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.00000 0.5617 0.6628 0.6430 0.7489 1.0000
Modality 4 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.000000 0.000000 0.04486 0.000000 1.00000
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.00000 0.00000 0.2478 0.3440 0.6344 1.0000

--- Summary for K = 8 ---
Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.7533 0.8372 0.8478 0.8463 0.8578 0.8911

beta.ar:
Modality 1 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
0.000000 0.003889 0.015556 0.072820 0.047778 0.885556
Modality 2 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
Modality 3 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
Modality 4 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1
Modality 5 :
  Min. 1st Qu.  Median      Mean 3rd Qu.  Max.
    1         1         1         1         1         1

```

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gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.004444 0.016667 0.044890 0.047778 0.505000
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.000000 0.000000 0.01419 0.000000 1.00000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000000 0.0000000 0.0000000 0.0000639 0.0000000 0.7433333
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000000 0.0000000 0.0000000 0.001544 0.0000000 1.000000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.000000 0.000000 0.01349 0.000000 1.00000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.1244 0.1506 0.1611 0.1654 0.1767 0.2667

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.003333 0.017778 0.030487 0.044444 0.677778
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.004444 0.020000 0.034290 0.051111 0.423333
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.000000 0.000000 0.04228 0.000000 1.00000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000 0.7072 0.8256 0.8108 0.9594 1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.000000 0.000000 0.02219 0.000000 0.86000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.

```

```
0.00000 0.00000 0.00000 0.09190 0.07333 1.00000
', '\n`~`'))
```

```
## ‘‘‘
##
##
## --- Summary for K = 1 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.4678  0.5061  0.5139  0.5139  0.5217  0.5494
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.0000  0.0950  0.1394  0.1359  0.1711  0.8944
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.00000 0.09778 0.14222 0.13881 0.17444 0.72889
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.000000 0.005000 0.008333 0.102034 0.031111 1.000000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.0005556 0.0050000 0.0083333 0.0780456 0.0238889 1.0000000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.000000 0.006111 0.015000 0.162453 0.117778 1.000000
##
## --- Summary for K = 2 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.2128  0.2356  0.2428  0.2427  0.2506  0.2733
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
```



```

## 0.00000 0.05278 0.10278 0.10558 0.14222 0.80556
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.00000 0.05556 0.10667 0.10858 0.14667 0.66611
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.000000 0.001667 0.060556 0.416370 1.000000 1.000000
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.000000 0.001111 0.005000 0.250076 0.386250 1.000000
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000 0.1328 1.0000 0.6989 1.0000 1.0000
##
## --- Summary for K = 3 ---
## Z.ar:
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.1217 0.1522 0.1606 0.1621 0.1706 0.2417
##
## beta.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.00000 0.01944 0.05833 0.06923 0.10222 0.89278
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:

```

```

## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.02000 0.06111 0.07181 0.10667 0.62556
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.7661  1.0000  0.7733  1.0000  1.0000
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000000 0.0000000 0.0005556 0.2259396 0.2123611 1.0000000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.1172  1.0000  0.6837  1.0000  1.0000
##
## --- Summary for K = 4 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.03833 0.06111 0.06833 0.06936 0.07556 0.14000
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01944 0.06000 0.06899 0.10000 0.84833
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.02111 0.06444 0.07269 0.10667 0.65500
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.5683  0.8311  0.7057  1.0000  1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.2704  0.6675  1.0000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.9950  1.0000  0.8584  1.0000  1.0000

```

```

##
## --- Summary for K = 5 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.03111 0.04944 0.05611 0.05862 0.06556 0.14222
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01556 0.04944 0.06017 0.08778 0.83556
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01667 0.05333 0.06408 0.09444 0.59889
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.2744  0.6322  0.5835  0.9906  1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.8622  1.0000  1.0000  1.0000  1.0000  1.0000
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.2524  0.5406  1.0000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.8672  1.0000  0.8440  1.0000  1.0000
##
## --- Summary for K = 6 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.03667 0.05944 0.06889 0.07106 0.07944 0.15111
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.005556 0.028889 0.045548 0.067222 0.799444
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

```

```

##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.005556 0.031111 0.048393 0.073333 0.595556
##   Modality 2 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.000000 0.08778 0.36773 0.81778 1.00000
##   Modality 3 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.6656 1.0000 1.0000 0.9996 1.0000 1.0000
##   Modality 4 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0000 0.0000 0.1860 0.2053 1.0000
##   Modality 5 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.4906 0.8444 0.6852 0.9828 1.0000
##
## --- Summary for K = 7 ---
## Z.ar:
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.2239 0.2517 0.2622 0.2641 0.2744 0.3517
##
## beta.ar:
##   Modality 1 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.00500 0.01778 0.03216 0.04222 0.74000
##   Modality 2 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.005556 0.020000 0.033246 0.045556 0.497778
##   Modality 2 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.000000 0.00000 0.13399 0.08167 1.00000

```

```

## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.5617 0.6628 0.6430 0.7489 1.0000
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.04486 0.00000 1.00000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0000 0.2478 0.3440 0.6344 1.0000
##
## --- Summary for K = 8 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.7533 0.8372 0.8478 0.8463 0.8578 0.8911
##
## beta.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.003889 0.015556 0.072820 0.047778 0.885556
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.004444 0.016667 0.044890 0.047778 0.505000
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.01419 0.00000 1.00000
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000000 0.0000000 0.0000000 0.0000639 0.0000000 0.7433333
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.000000 0.000000 0.001544 0.000000 1.000000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.01349 0.00000 1.00000
##
## --- Summary for K = 9 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.1244 0.1506 0.1611 0.1654 0.1767 0.2667
##

```

```

## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.000000 0.003333 0.017778 0.030487 0.044444 0.677778
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.000000 0.004444 0.020000 0.034290 0.051111 0.423333
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.04228 0.00000 1.00000
##   Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.0000 0.7072 0.8256 0.8108 0.9594 1.0000
##   Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.02219 0.00000 0.86000
##   Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.09190 0.07333 1.00000
##
## ‘‘‘

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1–9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.